CS 457 – Lecture 17
Global Internet

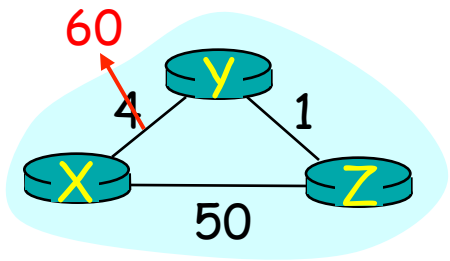
Fall 2011

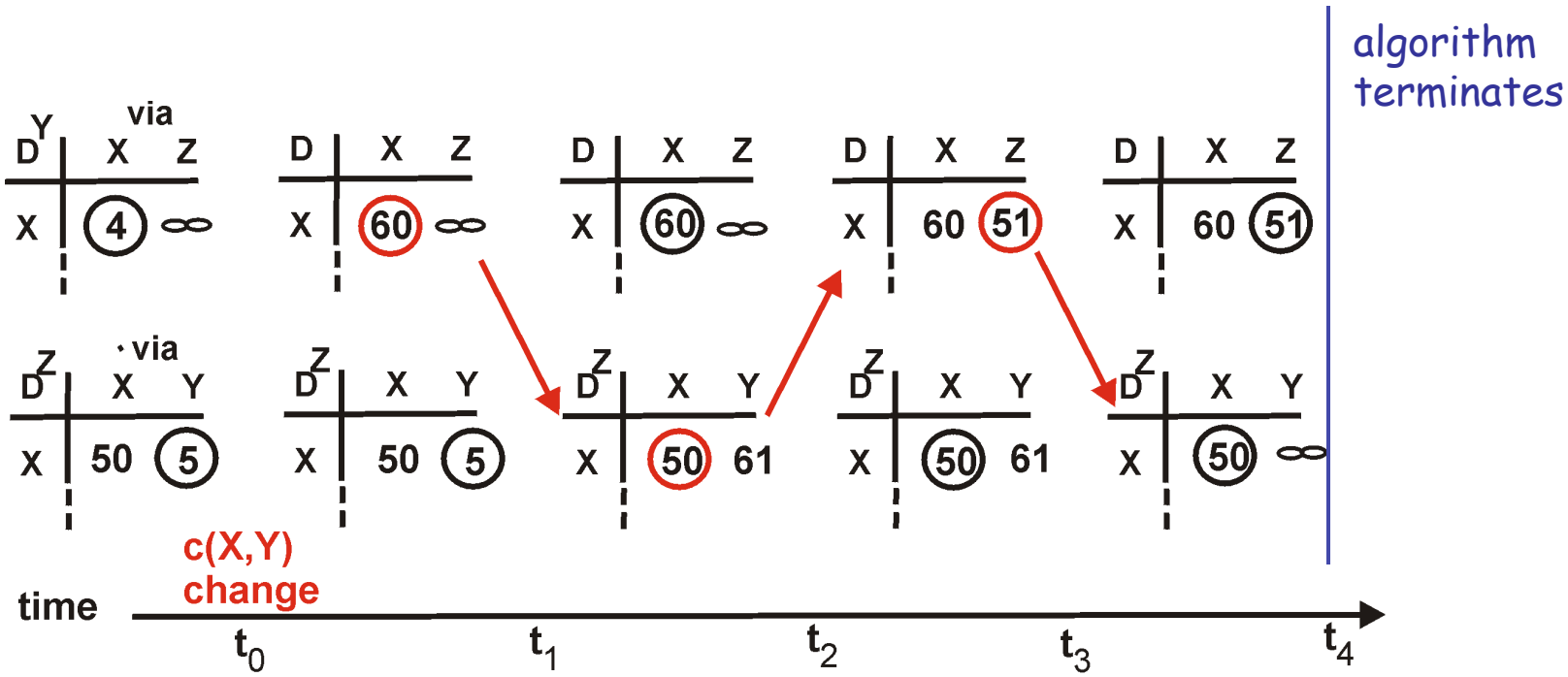
Distance Vector: Poison Reverse

If Z routes through Y to get to X :

Z tells Y its (Z's) distance to X is infinite (so Y won't route to X via Z)

Still, can have problems when more than 2 routers are involved





Comparison of LS and DV algorithms

Message complexity

- LS: with n nodes, E links, O(nE) messages sent
- DV: exchange between neighbors only
  - Convergence time varies

Speed of Convergence

- LS: $O(n^2)$ algorithm requires O(nE) messages
- DV: convergence time varies
  - May be routing loops
  - Count-to-infinity problem

Robustness: what happens if router malfunctions?

LS:

- Node can advertise incorrect *link* cost
- Each node computes only its *own* table

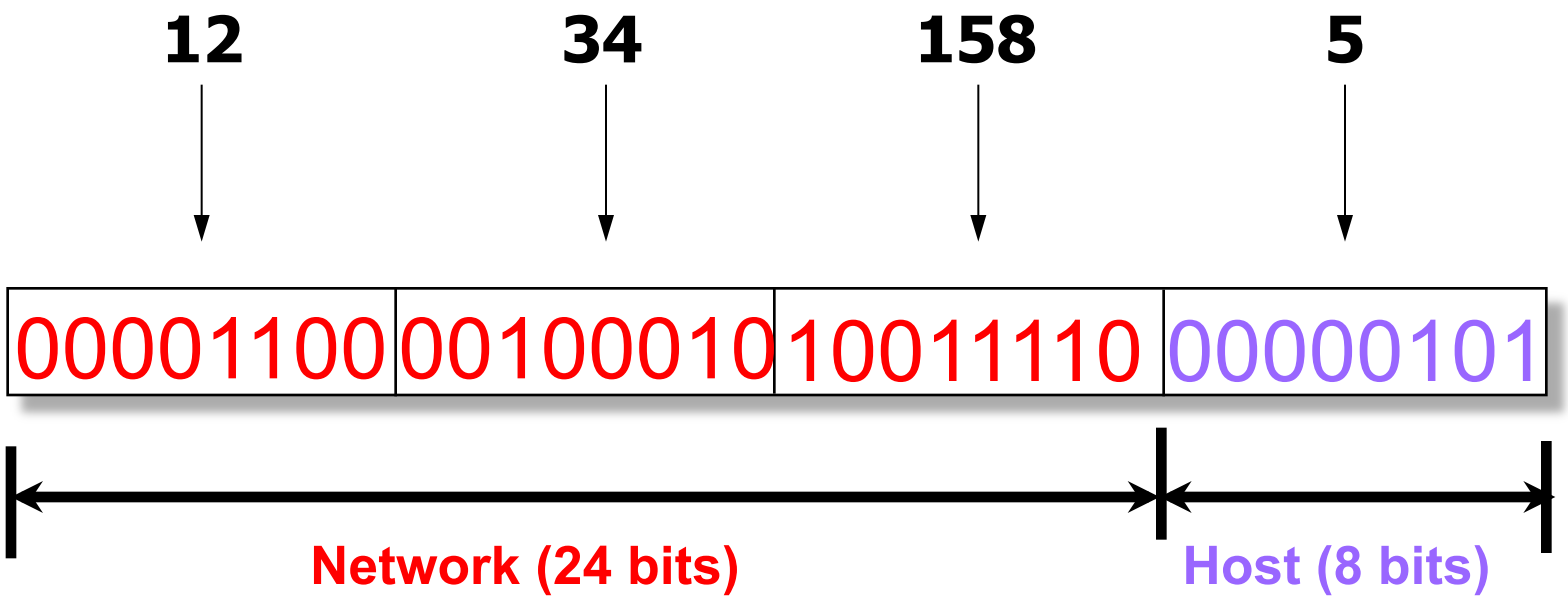
DV:

- DV node can advertise incorrect *path* cost
- Each node's table used by others (error propagates)

Address Allocation

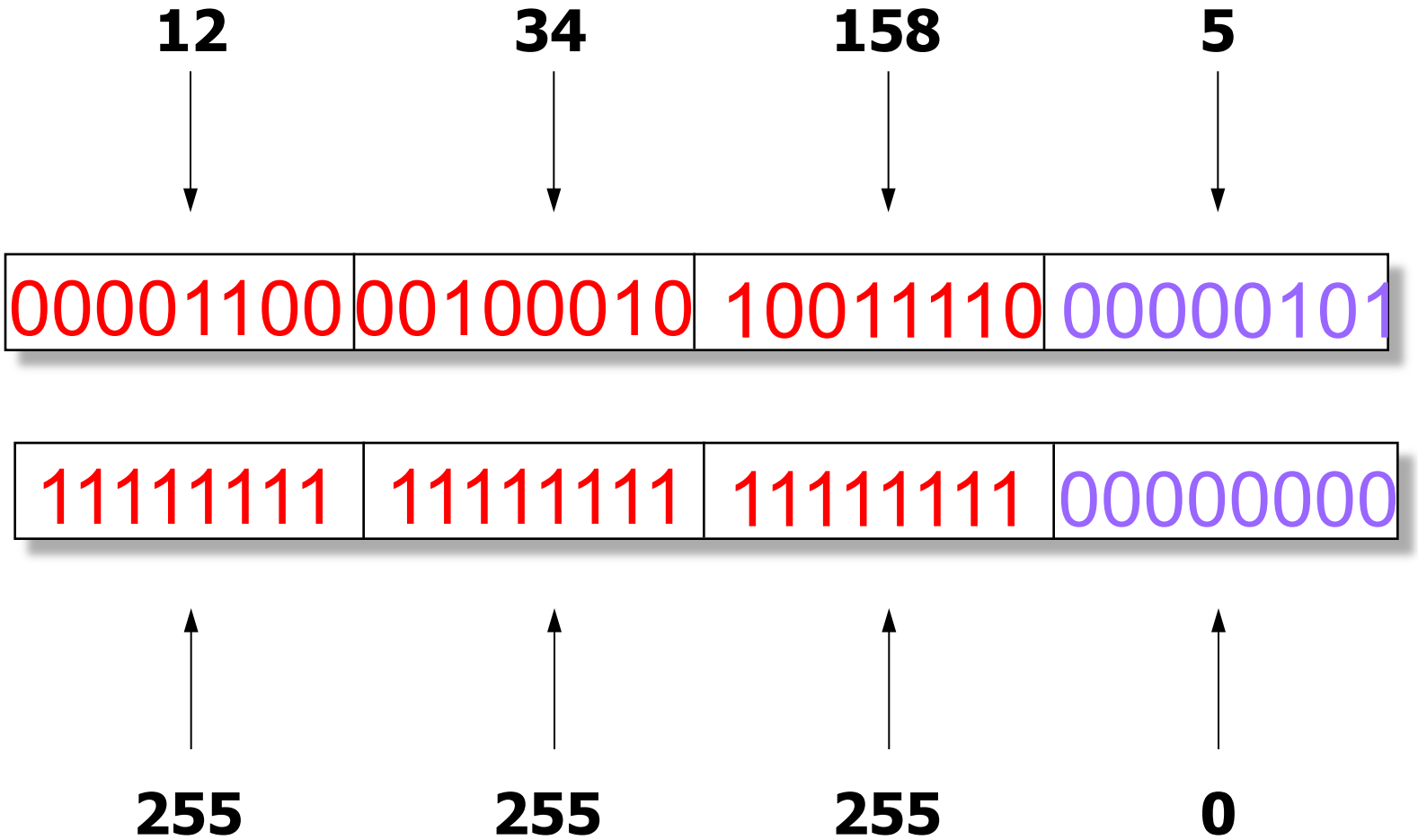
Hierarchical Addressing: IP Prefixes

- Divided into network & host portions (left and right)
- 12.34.158.0/24 is a 24-bit prefix with 2^8 addresses



IP Address and 24-bit Subnet Mask

Address



Mask

Classful Addressing

- In the olden days, only fixed allocation sizes
  - Class A: 0*
    - Very large /8 blocks (e.g., MIT has 18.0.0.0/8)
  - Class B: 10*
    - Large /16 blocks (e.g,. Princeton has 128.112.0.0/16)
  - Class C: 110*
    - Small /24 blocks (e.g., AT&T Labs has 192.20.225.0/24)
  - Class D: 1110*
    - Multicast groups
  - Class E: 11110*
    - Reserved for future use
- This is why folks use dotted-quad notation!

Classless Inter-Domain Routing (CIDR)

Use two 32-bit numbers to represent a network.
Network number = IP address + Mask

IP Address : 12.4.0.0 **IP Mask: 255.254.0.0**

Address	00001100	00000100	00000000	00000000
Mask	11111111	11111110	00000000	00000000

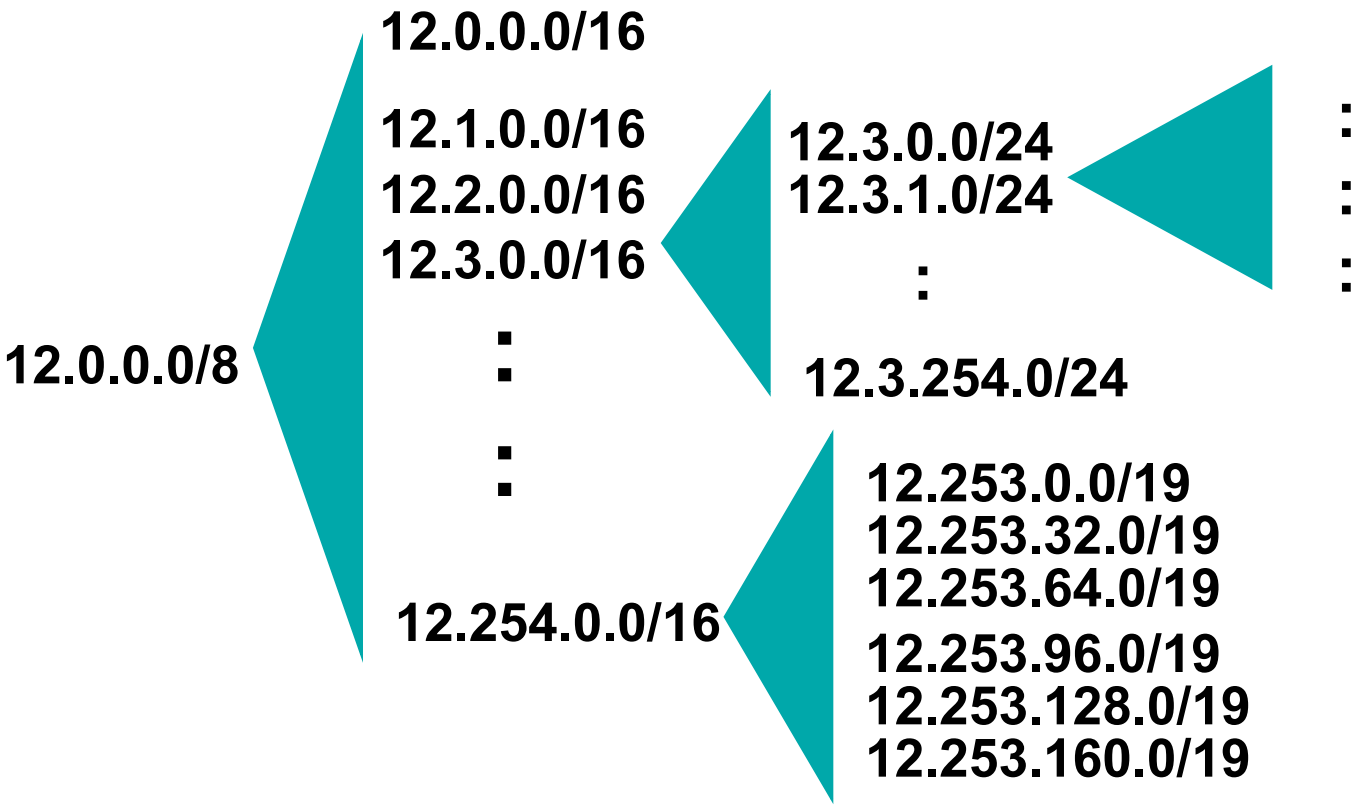
Network Prefix **for hosts**

Written as 12.4.0.0/15

CIDR: Hierarchal Address Allocation

Prefixes are key to Internet scalability

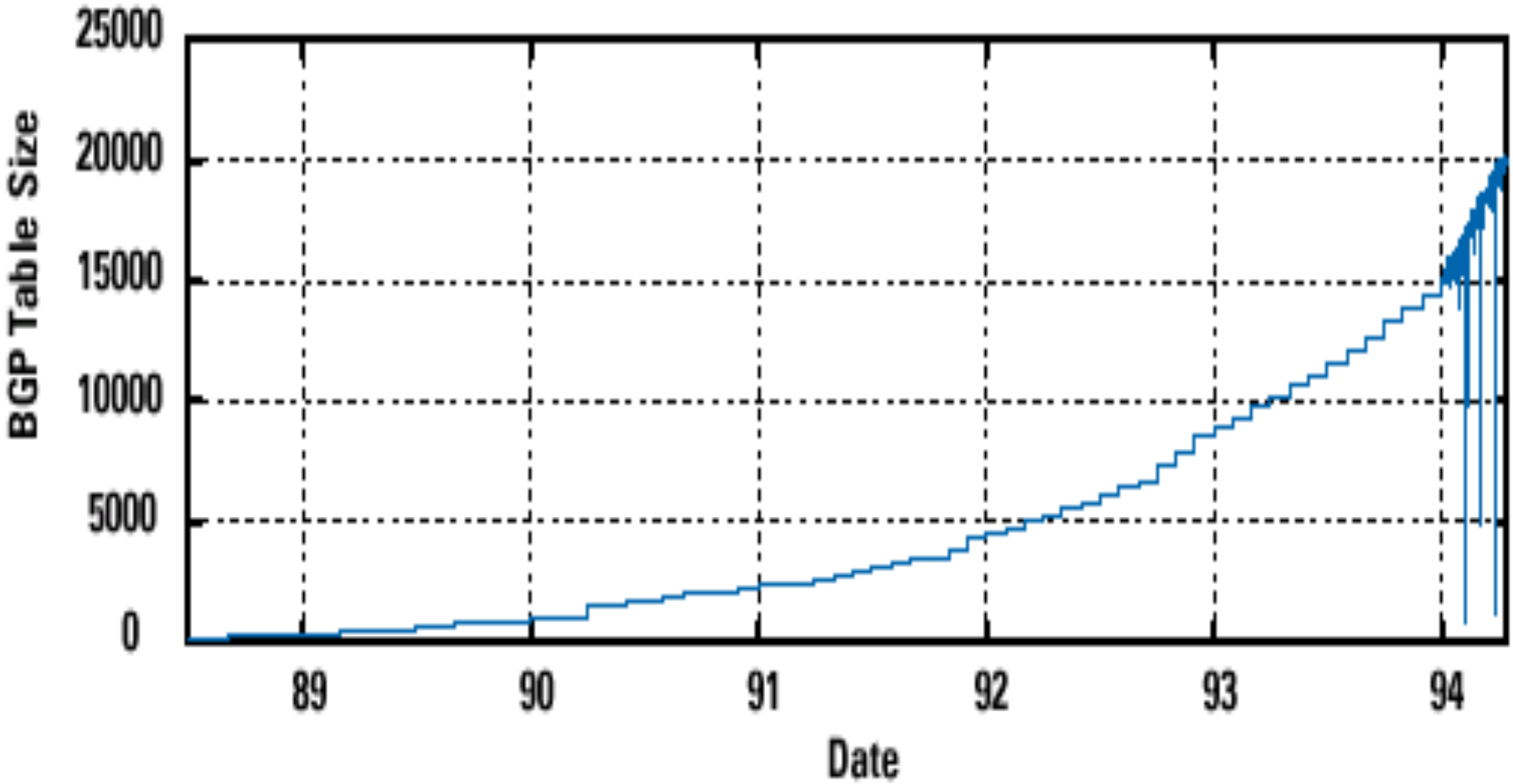
- Address allocated in contiguous chunks (prefixes)
- Routing protocols and packet forwarding based on prefixes
- Today, routing tables contain ~250,000-300,00 prefixes



Scalability Through Hierarchy

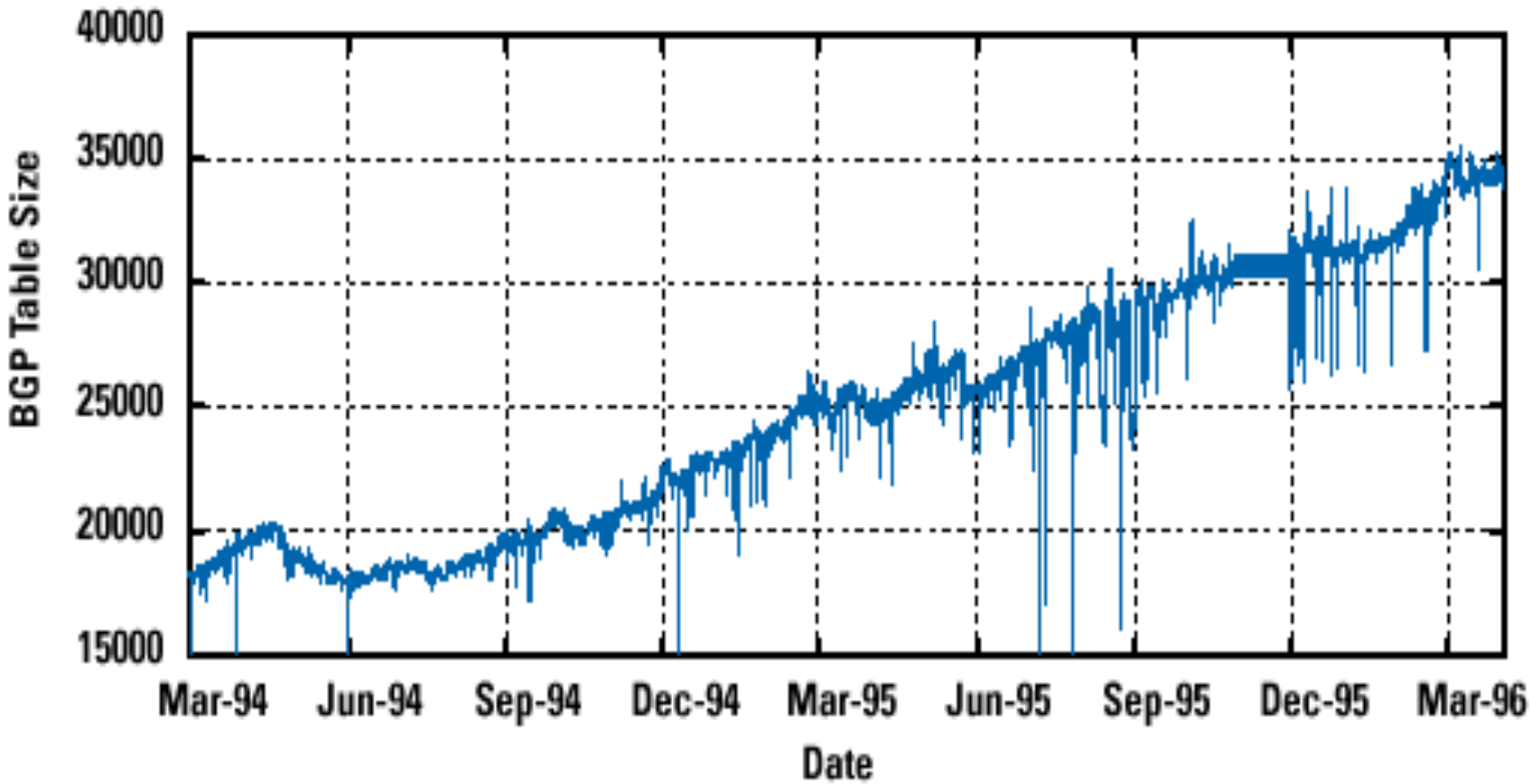
- Hierarchical addressing
  - Critical for scalable system
  - Don’t require everyone to know everyone else
  - Reduces amount of updating when something changes
- Non-uniform hierarchy
  - Useful for heterogeneous networks of different sizes
  - Initial class-based addressing was far too coarse
  - Classless Inter Domain Routing (CIDR) helps
- Next few slides
  - History of the number of globally-visible prefixes
  - Plots are # of prefixes vs. time

Pre-CIDR (1988-1994): Steep Growth



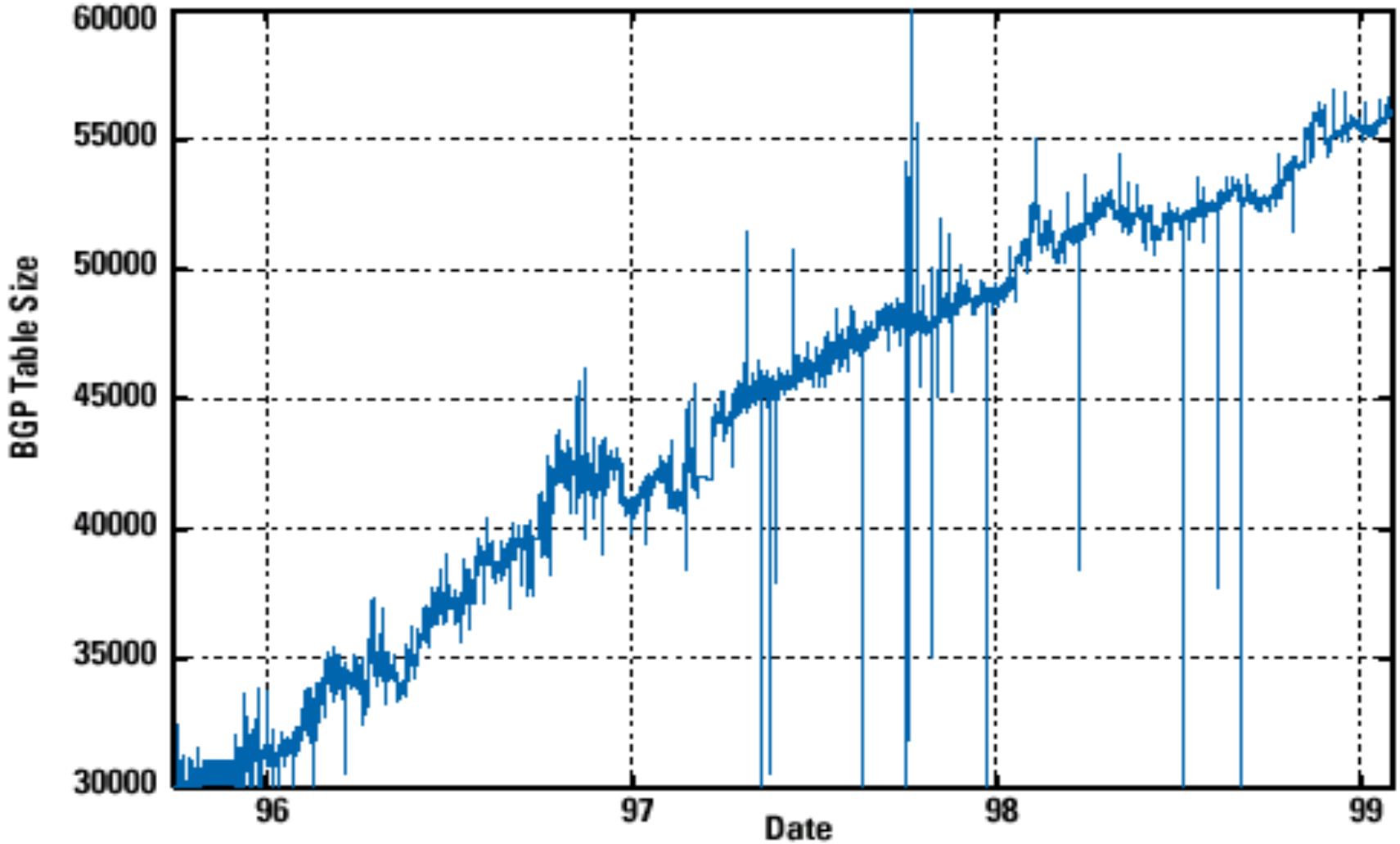
Growth faster than improvements in equipment capability

CIDR Deployed (1994-1996): Much Flatter



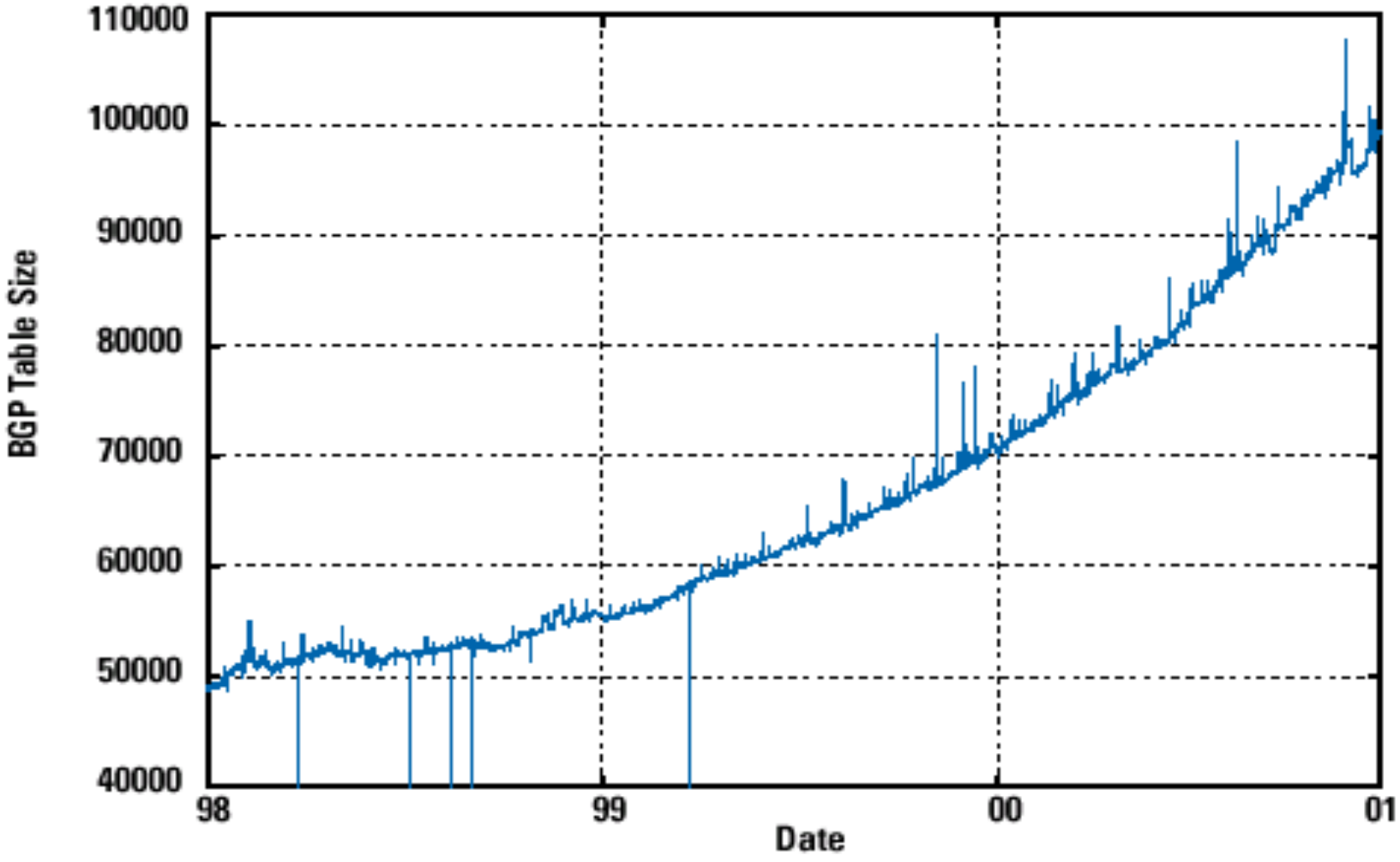
Efforts to aggregate (even decreases after IETF meetings!)

CIDR Growth (1996-1998): Roughly Linear



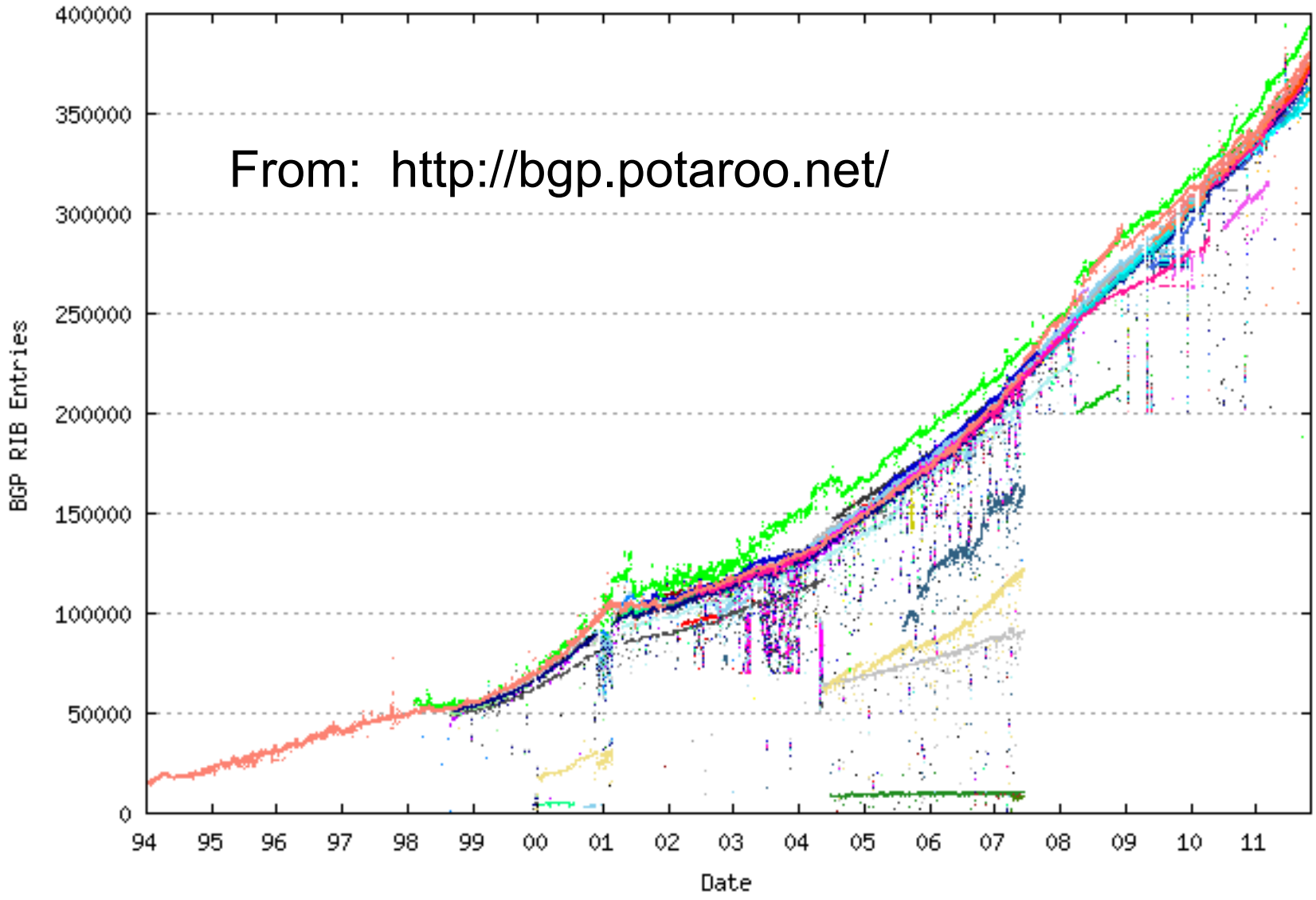
Good use of aggregation, and peer pressure in CIDR report

Boom Period (1998-2001): Steep Growth



Internet boom and increased multi-homing

Long-Term View (1989-2011)



Obtaining a Block of Addresses

- Separation of control
  - Prefix: assigned *to* an institution
  - Addresses: assigned *by* the institution to their nodes
- Who assigns prefixes?
  - Internet Corporation for Assigned Names and Numbers
    - Allocates large address blocks to Regional Internet Registries
  - Regional Internet Registries (RIRs)
    - E.g., ARIN (American Registry for Internet Numbers)
    - Allocates address blocks within their regions
    - Allocated to Internet Service Providers and large institutions
  - Internet Service Providers (ISPs)
    - Allocate address blocks to their customers
    - Who may, in turn, allocate to their customers…

Figuring Out Who Owns an Address

- Address registries
  - Public record of address allocations
  - Internet Service Providers (ISPs) should update when giving addresses to customers
  - However, records are notoriously out-of-date
- Ways to query
  - UNIX: “whois –h whois.arin.net 128.112.136.35”
  - http://www.arin.net/whois/
  - http://www.geektools.com/whois.php
  - …

Example Output for 128.112.136.35

OrgName: Princeton University
OrgID: PRNU
Address: Office of Information Technology
Address: 87 Prospect Avenue
City: Princeton
StateProv: NJ
PostalCode: 08544-2007
Country: US
NetRange: 128.112.0.0 - 128.112.255.255
CIDR: 128.112.0.0/16
NetName: PRINCETON
NetHandle: NET-128-112-0-0-1
Parent: NET-128-0-0-0-0
NetType: Direct Allocation
RegDate: 1986-02-24

Hard Policy Questions

- How much address space per geographic region?
  - Equal amount per country?
  - Proportional to the population?
  - What about addresses already allocated?
- Address space portability?
  - Keep your address block when you change providers?
  - Pro: avoid having to renumber your equipment
  - Con: reduces the effectiveness of address aggregation
- Keeping the address registries up to date?
  - What about mergers and acquisitions?
  - Delegation of address blocks to customers?
  - As a result, the registries are horribly out of date
- Many Of These Questions Still Being Answered
  - Let's understand how Internet routing works…

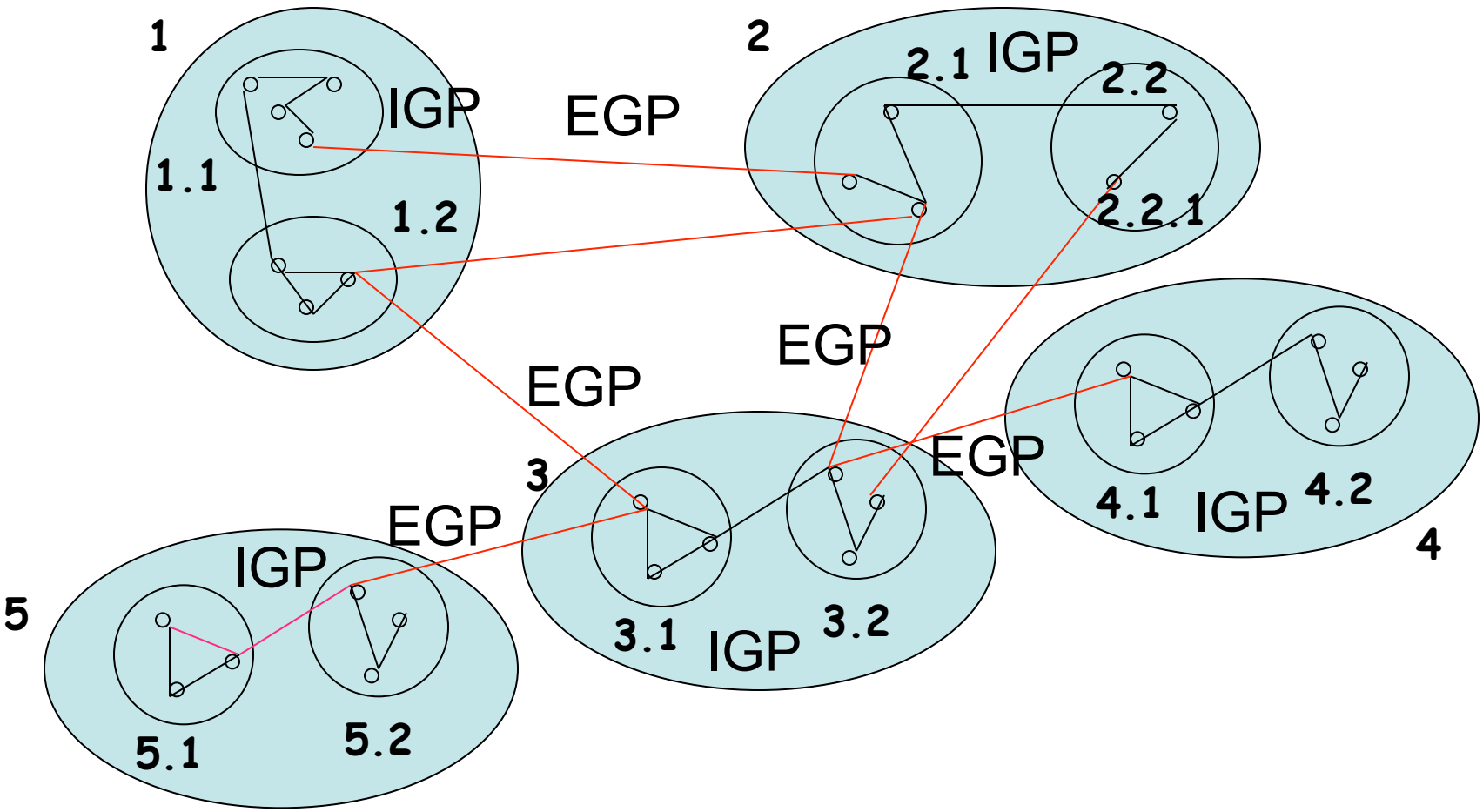
Global Internet Routing

- Objective is to provide routes to prefixes
  - Could be an IPv4 prefix
  - Could be an IPv6 prefix
- Route should get you to the right "***??***"
  - Route to 129.82.0.0/16 should get you to ColoState
  - Once here, packet may follow a RIP or OSPF route to the right subnet

Autonomous Systems

- What is an AS?
  - a set of routers under a single technical administration
  - uses an *interior gateway protocol (IGP)* and common metrics to route packets within the AS
  - uses an *exterior gateway protocol (EGP)* to route packets to other AS's
- AS may use multiple IGPs and metrics, but appears as single AS to other AS's

Example


1
IGP
1.1
1.2
EGP
2
2.1
IGP
2.2
2.2.1
EGP
EGP
EGP
3
3.1
IGP
3.2
EGP
4.1
IGP
4.2
4
EGP
IGP
5
5.1
5.2

BGP Routing Choices

- Link state or distance vector?
  - no universal metric - policy decisions
- Problems with distance-vector:
  - Bellman-Ford algorithm slow to converge (counting to infinity problem)
- Problems with link state:
  - metric used by routers not the same - loops
  - LS database too large - entire Internet
  - may expose policies to other AS's

What's Next

- Read Chapter 1, 2, 3, and 4.1-4.3
- Next Lecture Topics from Chapter 4.4 - 4.6
  - Multicast, MPLS, and Routing Wrap-up
- Homework
  - Due ***Thursday*** in lecture
- Project 2
  - Due Friday